IMPLEMENTATION HANDOFF SPECIFICATION

Trapp Fantasy Studio - UI & Usability Enhancements

Actionable component, layout, and interaction blueprint for coding agents (100% Feature-Preserving)

Prototype: docs/mockups/usability-improvements/ | Contracts: docs/feature-preservation.md | Reductions: 0 Approved

MANDATORY PRESERVATION GUARDRAILS FOR IMPLEMENTATION AGENTS

1. Zero Feature Reductions: Every control, calculation, caveat, tooltip, keyboard path, and export in TASK-1..TASK-146 must be preserved. docs/features/approved-changes.md contains zero exceptions.
2. Do Not Change Ranking Exports (TASK-3, TASK-17, TASK-30): Workspace CSS must never bleed into src/export.css or RankingPreview. Original (1304px) and Compact (720px) PNG downloads must remain pixel-identical to baselines.

1. Design System & Semantic Tier Tokens

`--bg / --panel`	`#F3F5FB / #FFFFFF`	Quiet workspace background and primary white card surfaces.
`--ink / --muted`	`#17223B / #526078`	Sticky dark topbar, primary text (14px), and muted labels (12px, AA).
`--accent / --tint`	`#3949AB / #EEF1FF`	Primary buttons, active tabs, focus rings, and selected-row fills.
`--good / --warn / --bad`	`#176B50 #B55D14 #9E3D40`	Positive/ready states, bye/uncertainty alerts, and live/negative badges.
`--tier-1 .. --tier-4`	`#DBEAFE #D1FAE5 #FEF3C7 #FEE2E2`	Restored 4-tier Position Heatmap (#1-2 Blue, #3-4 Green, #5 Amber, #6+ Red).

2. Component-by-Component UI & Usability Specifications

2.1 Shared App Shell, Unified Topbar & Page Command Bar (AppShell.tsx)

Preserved Contracts: TASK-27, TASK-39, TASK-104, TASK-120, TASK-146

* Fix Live Matchups Account Menu Bug: In live-1440.png, hiding <header> causes .account-menu (Alex Morgan) to drop into the light workspace in white-on-white text overlapping Matchup Slot 2. Keep Brand, the 7 primary nav buttons, and .account-menu inside one sticky dark header.app-topbar (min-height: 56px; 44px on Live Matchups).
* Reduce Vertical Chrome by 95px: Merge the standalone page heading, League/Season/Week selects, freshness pill, and page-level actions into a single horizontal section.page-command-bar.
* 1-Click Sub-Tool Switcher: Render nav.subtool-switcher at the top of Other tools, Players, and Trade analyzer so users can switch directly between Players and Trade Analyzer in 1 click while keeping Other tools active in the 7-item main navigation.

2.2 Rankings Studio & Isolated Export Preview (RankingEditor.tsx, RankingPreview.tsx)

Preserved Contracts: TASK-3, TASK-13..13.3, TASK-15..17, TASK-19, TASK-23, TASK-30, TASK-33, TASK-63, TASK-83, TASK-86

* Top Studio Toolbar (.studio-toolbar): Move Save rankings, Undo move, Draft & save status, Copy edition, Revision history, and Keyboard shortcuts to the top of the left editor pane instead of burying them below the team list.
* Progress Bar & Inline Take Previews: Add an Edition completion pill (X of N takes ready) with a progress bar. In each .rankrow, display a drag handle (::), circular rank badge, movement badge (^1 / v1), a 1-line truncated preview of the team take, and a [OK] Take ready vs Draft needed badge alongside Up/Down buttons.
* Sticky Right-Hand Take Studio: Make aside.aside sticky (top: 68px) and add <- Prev / Next #N -> stepper buttons so commissioners can write takes from Rank #1 to #N sequentially without scrolling back and forth.
* Dual Export Preview Toggle: In Preview & export, add a toggle to preview both Desktop Original (1304px) and Mobile Compact (720px) layouts inside the isolated preview host while keeping src/export.css untouched.

2.3 My Weekly Overview & Proactive Advisor Previews (WeeklyOverviewPage.tsx)

Preserved Contracts: TASK-90, TASK-91, TASK-92, TASK-95

* Win-Probability Tug-of-War Bar (.tug-bar-wrap): Render a two-sided visual win-probability bar under each league matchup header with explicit uncalibrated estimate labels.
* Actionable Impact Badges on Buttons: Display live calculated gains directly on the advisor buttons: Lineup advisor (+17.8 pts), Waiver advisor (2 upgrades), and Dismissed alerts (N).
* Inline Quick-Preview Drawer (.inline-advisor-box): Add a collapsible Quick preview table directly inside each league card showing the top starter swaps and waiver upgrades at a glance, while retaining the full interactive Lineup & Waiver Advisor modal.

2.4 Live Matchups 4-Box Board (LiveMatchupsPage.tsx)

Preserved Contracts: TASK-106, TASK-112, TASK-146

* Managed Team Markers: Prefix the user managed teams in the first four default games with [*] in the left sidebar game picker (.live-game-picker).
* Quarter Header Win Probability Pill: Display the uncalibrated win estimate badge in each .quarter-heading beside the remove (x) button.
* Live NFL Status Column (td.live-status-col): Add a compact status column (LIVE Q3, FINAL, SUN 3:25, BYE) inside table.dense-roster-table while preserving 18-player no-scroll desktop fit at 1440x900 and 2048x942, independent quarter scrolling at 1280x720, and full page flow on 390px phones.

2.5 Season Insights & Restored Position Heatmap (InsightsPage.tsx, SeasonStrength.tsx)

Preserved Contracts: TASK-4, TASK-5, TASK-28, TASK-29, TASK-40, TASK-50..52, TASK-108, TASK-109, TASK-111, TASK-113

* Persistent Selected Team Evidence Aside: Restore the two-column .split layout across all 7 Season Insights views (Team scoring, Weekly trends, Roster strength, Remaining schedule, Manager report card, Trades, Pickups) so clicking a team updates the right-hand Selected Team Inspector without hijacking the active tab, plus an explicit Open in Weekly trends -> button.
* Restored 4-Tier Semantic Position Heatmap: Style position group rank cells with .tier-1 (ranks 1-2 sky blue), .tier-2 (ranks 3-4 emerald green), .tier-3 (mid amber), and .tier-4 (bottom coral red) across all league-configured positions (QB, RB, WR, TE, K, DEF), showing both #Rank and Points as in docs/references/season-insights/position-group-rankings.png.
* Stacked Roster Projections X-Axis: Include the 0 to 2200 horizontal axis scale below the stacked positional bars to match docs/references/season-insights/roster-projections.png.

2.6 Playoff Simulator & Histogram Mode Toggle (PlayoffForecast.tsx, PlayoffCalibration.tsx)

Preserved Contracts: TASK-34, TASK-36, TASK-41, TASK-42, TASK-55..57, TASK-69, TASK-73..81, TASK-116

* Segmented Table / Histogram Toggle (TASK-69): Add a visible toggle in the Forecast table header switching between the round-by-round probability table (Make playoffs, Earn bye, Reach semifinal, Reach final, Win title) and the simulated win-distribution / seed probability Histogram view (.histogram-grid).
* Restored Selected Team Path Inspector: Keep the selected team record, projected wins, round odds, and bracket tiebreaker rules pinned in the right-hand aside panel, with Weekly lineup projection coverage expandable below the table.
* Advanced Calibration Suite: Preserve the Advanced tab with all 5 sub-views (Convergence, Weekly diagnostics, Season comparisons, Reliability, Final seeding), CSV download, and full replay JSON export.

2.7 Players Matrix & Trade Exchange Banner (PlayerProfilesPage.tsx, TradeAnalyzerPage.tsx)

Preserved Contracts: TASK-93, TASK-94, TASK-99, TASK-102, TASK-103, TASK-110, TASK-115, TASK-117, TASK-118

* Side-by-Side Weekly Comparison Matrix (Players): When 2+ players are selected for comparison, render a unified comparison table at the top of Selected Players showing window average, window total, combined + provider-split projections, and week-by-week observed scores side by side above the individual player profile cards.
* Unified Trade Exchange Banner (.trade-exchange-banner): At the top of Build a trade, render a 3-column summary banner (Your Team Sends <-> Partner Sends) showing selected assets, before-to-after legal starting lineup scores, net point deltas, and positional depth changes in one bar. Retain the conditional open-slot confirmation for uneven trades (TASK-102).

2.8 League History Explorer & Profile/League Safeguards (HistoryPage.tsx, ProfilePage.tsx)

Preserved Contracts: TASK-6..8, TASK-20, TASK-70, TASK-89, TASK-97, TASK-101, TASK-122..145

* Auto-Populated Record Evidence Inspector: On initial mount or view switch in HistoryPage.tsx, auto-select the top rivalry/record row so the right-hand Record evidence panel displays real game provenance immediately instead of an empty prompt.
* High-Contrast Fantasy Wrapped Cards: Present Wrapped (Personal recap and League recap) in a 4-metric story grid with explicit consent, immutable snapshot creation, and revocation controls.
* Profile & League Connections: Retain verified email & Auth0 identity display, password recovery flow, Server-Managed Codex readiness vs personal API key precedence, private ESPN cookie return flow, and active/archived league management.

3. Mandatory Verification Gates Before Completing Implementation

1. Reference Mockup Inspection: Open docs/mockups/usability-improvements/index.html (and styles.css / app.js) alongside docs/feature-preservation.md.
2. Unit, Component & Calculation Suite: Run `npm test -- --maxWorkers=4` and verify all suites pass without weakening any existing assertions.
3. Browser & Responsive Checks: Run `npx playwright test --workers=2` across 1440x900, 2048x942, 1280x720, 390x844, and 320x568 with zero overflow.
4. Ranking PNG Baseline Parity: Confirm downloaded Original (1304px) and Compact (720px) ranking PNGs have zero changed pixels against baselines.
5. Static Quality Gates: Confirm `npm run typecheck`, `npm run lint`, `npm run format:check`, and `npm run build` succeed with zero errors.